

1. A partial differential equation requires
 - A. exactly one independent variable
 - B. more than one dependent variable
 - C. two or more independent variables
 - D. equal number of dependent and independent variable

2. If $u = \frac{\sqrt{xy}}{\sqrt{x} + \sqrt{y}}$, then the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ is
 - A. $\frac{1}{2} \frac{\sqrt{xy}}{\sqrt{x} + \sqrt{y}}$
 - B. $\frac{\sqrt{xy}}{\sqrt{x} + \sqrt{y}}$
 - C. 0
 - D. 1

3. Maximum value of the function $x^4 + y^4 - x^2 - y^2 + 1$
 - A. 0
 - B. 1
 - C. 2
 - D. 3

4. A function $f(x, y)$ attains maximum at a point (a, b) if
 - A. $r - s^2 > 0, r < 0$
 - B. $r - s^2 > 0, r > 0$
 - C. $r - s^2 > 0$
 - D. $r - s^2 < 0$

5. For a non singular matrix A if PAQ is in normal form A^{-1} is equal to
 - A. PQ
 - B. QP

- C. $P + Q$
- D. $Q - P$

6. If $W = xy + z$, $x = \cos t$, $y = \sin t$, $z = t$, then $\frac{dW}{dt}$ is

- A. $1 - \sin 2t$
- B. $1 + \cos 2t$
- C. $1 - \cos t$
- D. $\cos 2t$

7. The value of $\lim_{(x,y) \rightarrow (0,0)} \frac{2x^2y}{x^4 + y^2}$ is

- A. 1
- B. 2
- C. 0
- D. does not exist

8. The degree of $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$ is

- A. 0
- B. 1
- C. 2
- D. 3

9. If $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, evaluate $\frac{\partial(x, y, z)}{\partial(r, \theta, z)}$ is

- A. r^3
- B. r^2
- C. r
- D. 1

10. Taylor's series approximation to the function $x^2y - 2$ about the point $(1,-2)$ is

- A. $y - 4x + 2$
- B. $2x + y - 2$
- C. $y + 4x + 2$
- D. $x - 2y + 3$

11. Critical points of $x^3y^2(1 - x - y)$ is

- A. $\left(\frac{1}{2}, \frac{1}{4}\right)$
- B. $(0,0), (1,2)$
- C. $(0,0), \left(\frac{1}{2}, \frac{1}{3}\right)$
- D. $(0,0), (1,1)$

12. If $u = e^x \sin y$, $v = e^x \cos y$ then $\frac{\partial(u,v)}{\partial(x,y)}$ is

- A. 0
- B. 1
- C. $-e^{2x}$
- D. e^{2x}

13. If u is a homogeneous function of order p , then u_x and u_y both are homogeneous functions of order

- A. p
- B. $p-1$
- C. $p+1$
- D. $p-2$

14. If $u = e^{x^2+y^2+z^2}$, then $\frac{\partial^3 u}{\partial x \partial y \partial z}$ is

- A. $8xyz u$

B. $2xyz$

C. $3xyz$

D. xyz

15. Maclaurine series of any function $f(x,y)$ is a Taylor series at point

A. $(1,0)$

B. $(-1,3)$

C. $(0,0)$

D. $(0,1)$

16. For the function $x^2 + y^2 + z^2$ subject to the condition $ax + by + cz = a + b + c$ the Lagrange's multiplier λ is given by

A. $\frac{-2(a+b+c)}{a^2+b^2+c^2}$

B. $\frac{-3(a+b+c)}{a^2+b^2+c^2}$

C. $\frac{a+b+c}{a^2+b^2+c^2}$

D. $\frac{2(a+b+c)}{a^2+b^2+c^2}$

17. The maximum point of $f(x,y) = 1 - x^2y^2$ is

A. $(1, 2)$

B. $(0, 0)$

C. $(1, -1)$

D. $(2, 3)$

18. If $u = f(r,s,t)$, $r = \frac{x}{y}$, $s = \frac{y}{z}$, $t = \frac{z}{x}$ then $xu_x + yu_y + zu_z$ is

A. 2

B. 1

C. 3

D. 0

Ans. Key

1. C

2. A

3. B

4. A

5. B

6. B

7. D

8. A

9. C

10. A

11. C

12. C

13. B

14. A

15. C

16. A

17. B

18. D